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ITCS216: Fundamentals of Database Systems (1/2026)

Faculty of Information and Communication Technology

In-class Exercise 03: **Developing ERD**

SECTION 1

Objectives:

- Understand the business rules given the selected processes for Tiny College
- Develop the ERD from given business rules.

Instructions:

1. Use the given business rules to develop the ERD for Tiny College.
2. Sketch the ERD using Chen Notations and submit by the end of the class.

Tiny College (TC) is divided into 4 schools which are school of business, school of arts & sciences, school of education, and school of applied sciences. Use the given 5 business rules to draw the ERD for TC:

R1: Each school can manage zero up to five departments.

R2: Each school contains unique name, location composed of the street, district, and province, and many phone numbers.

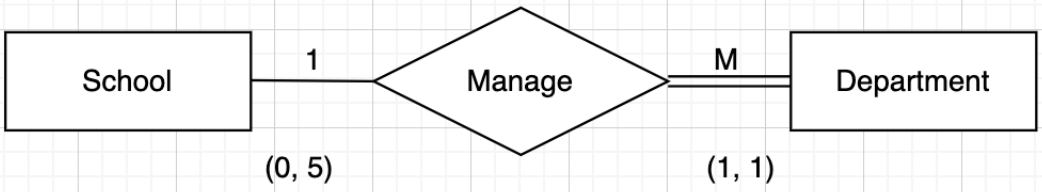
R3: Each school composes of many professors. In addition, each school must compose at least 1 professor and 10 professors at most.

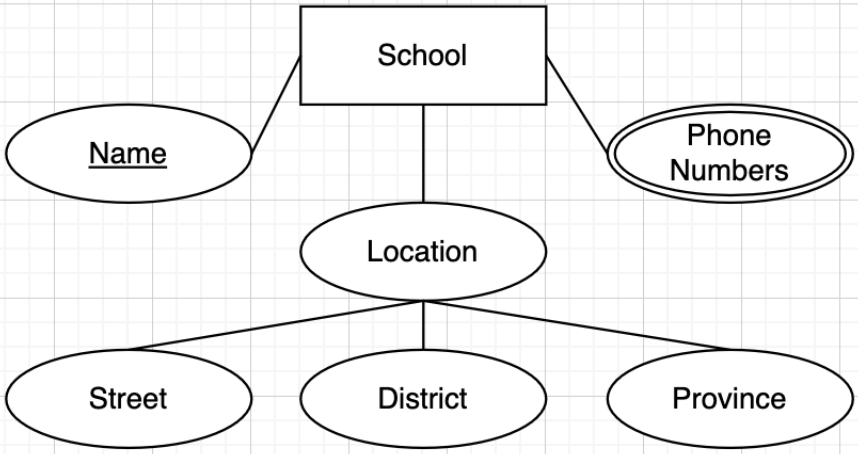
R4: Each school is administered by only one professor. This professor comprises voted date, and status which is marked to be the dean of the school.

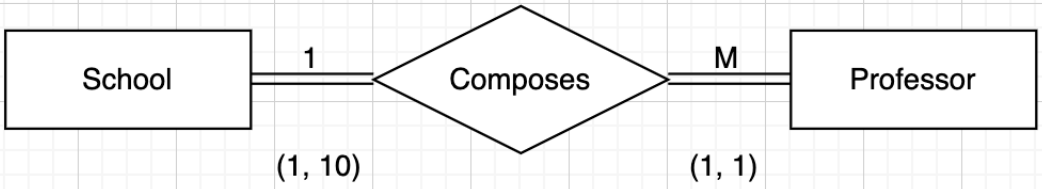
R5: A professor can be voted by many professors to be the dean of school.

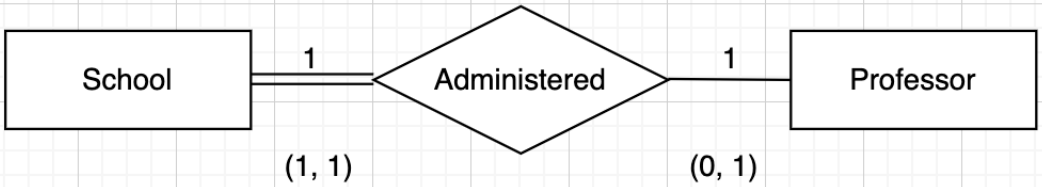
ERD

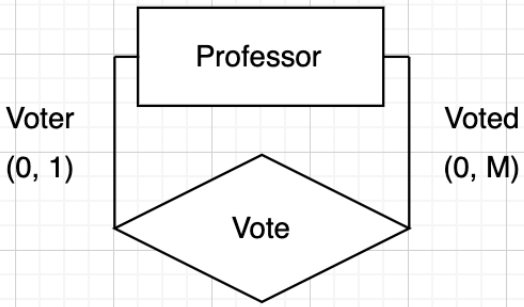
R1 to R5 by sequence, separated by line.


School
1
Manage
M
Department
(0, 5)
(1, 1)


School
Name
Phone Numbers
Location
Street
District
Province


School
1
Composes
M
Professor
(1, 10)
(1, 1)


School
1
Administered
1
Professor
(1, 1)
(0, 1)


Professor
Voter
(0, 1)
Voted
(0, M)
Vote